

Single-Selection Status Checkbox Documentation

Overview

This documentation describes a **single-selection (radio-like) checkbox system** implemented using a custom widget in Flutter. The component allows only **one status to be selected at a time**, controlled by a shared index variable.

This approach is useful when multiple status options are shown, but only one can be active (e.g., Asset Status, Filter Status, Lifecycle State).

State Variable

```
int selectedIndex = -1;
```

Purpose

- Stores the index of the currently selected status
 - Default value `-1` indicates **no selection**
 - Ensures only one checkbox is selected at any time
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Widget: `_buildStatusCheckbox`

Purpose

Creates a custom checkbox-like widget that: - Uses a rectangular outline - Displays a check icon when selected - Updates a shared selection index on tap

Source Code

```
Widget _buildStatusCheckbox(int index, String label) {  
  bool isChecked = selectedIndex == index;  
  
  return GestureDetector(  
    onTap: () {  
      setState(() {  
        selectedIndex = index;  
      });  
    },  
  ),  
);
```

```

child: Row(
  mainAxisAlignment: MainAxisAlignment.min,
  children: [
    Container(
      height: 20,
      width: 20,
      decoration: BoxDecoration(
        shape: BoxShape.rectangle,
        border: Border.all(
          color: isChecked ? AppColors.primary : AppColors.cBEBEBE,
          width: 1.5,
        ),
      ),
      child: isChecked
        ? Icon(
            Icons.check,
            color: AppColors.primary,
            size: 16,
          )
        : null,
    ),
    4.width, // spacing
    MyText(
      text: label,
      fontSize: 10.sp,
      color: AppColors.textColor,
    ),
  ],
),
);
}

```

Parameters

Parameter	Type	Description
<code>index</code>	int	Unique identifier for the checkbox
<code>label</code>	String	Text displayed next to the checkbox

Selection Logic

```
bool isChecked = selectedIndex == index;
```

- The checkbox is **checked** only if its index matches `selectedIndex`
- Tapping a checkbox updates `selectedIndex`
- Previously selected checkbox is automatically deselected

This mimics **radio-button behavior** while keeping a checkbox UI.

Interaction Handling

```
onTap: () {  
  setState(() {  
    selectedIndex = index;  
  });  
},
```

Behavior

- Entire row is tappable (better UX)
 - UI updates immediately via `setState`
 - Guarantees only one active status
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UI & Styling Details

Checkbox Container

- Size: `20 x 20`
- Shape: Rectangle
- Border:
 - Primary color when selected
 - Grey (`cBEBEBE`) when unselected
- Border width: `1.5`

Check Icon

- Icon: `Icons.check`
- Color: `AppColors.primary`
- Size: `16`
- Only rendered when selected

Label Text

- Widget: `MyText`
 - Font size: `10.sp`
 - Color: `AppColors.textColor`
-

Typical Usage Example

```
Wrap(  
  children: [  
    _buildStatusCheckbox(0, "Active"),  
    _buildStatusCheckbox(1, "Written-off"),  
    _buildStatusCheckbox(2, "De-registered"),  
    _buildStatusCheckbox(3, "Sold"),  
    _buildStatusCheckbox(4, "Out of Service"),  
  ],  
);
```

Advantages

- Enforces **single selection**
 - Lightweight and easy to maintain
 - Fully customizable UI
 - Better UX than default `Checkbox` for filters
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Limitations

- Selection cannot be toggled off (one option is always selected once tapped)
 - Uses `setState`, not suitable for large-scale apps without refactor
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Suggested Enhancements

- Allow deselection by toggling same index
 - Expose `onChanged(int index)` callback
 - Convert to reusable widget class
 - Integrate with Provider / Bloc / Riverpod
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Use Cases

- Status filters
 - Form step selection
 - Asset lifecycle state
 - Dashboard quick filters
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Summary

This implementation provides a **clean, radio-style checkbox solution** with full UI control and predictable selection behavior, ideal for small-to-medium Flutter screens requiring single-choice status selection.